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## **Maximizing Chemical Efficiency and Reducing Environmental Footprint with Rigless, Precision-Based Capillary Injection**

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### **Abstract**

Chemical injection remains essential in combating downhole issues such as corrosion, scale, and liquid loading within reservoir operations. However, traditional batch injection methods often suffer from inaccurate placement, excessive chemical consumption, and high operational costs. This study presents a novel rigless capillary injection methodology enabling precise and controlled chemical delivery to target zones, thereby enhancing treatment efficacy while minimizing chemical wastage and related expenditures. The primary goal is to optimize operational efficiency, reduce costs, and minimize environmental impact through highly targeted chemical placement.

The rigless capillary injection technique leverages an integrated approach combining samples analysis with advanced simulation-based modeling to optimize chemical dosing and accurate placement depth. The process begins with simulations to assess the precise chemical requirements and optimal depth for addressing treatment needs such as corrosion inhibition, scale control, and liquid loading mitigation. Based on these insights, the capillary string installation is tailored to ensure accurate placement within the wellbore. Additionally, a specially engineered injection valve is positioned at the production tubing intake to maintain chemical placement accuracy and prevent backflow.

Field trials of this rigless capillary injection approach have demonstrated substantial benefits, significantly improving chemical efficiency and cutting operational costs. Results show a 65% reduction in chemical usage per well, reflecting enhanced treatment accuracy and leading to substantial cost savings. Operating expenses were reduced by 52% through precise, targeted chemical delivery that eliminated the need for traditional cyclic injection cycles, which are often labor-intensive and prone to frequent adjustments. This minimized downtime, promoting stable, uninterrupted production. Moreover, the precision in chemical placement meant that less chemical volume was required, decreasing environmental impact by minimizing chemical waste and handling frequency. By directly addressing treatment points with minimal intervention, the rigless approach also reduced safety risks, limiting personnel exposure to hazardous chemicals. This analysis-driven, precision-based approach represents a significant advancement in sustainable reservoir management, aligning with industry goals for enhanced efficiency and adherence to environmental best practices.

The rigless capillary injection technology introduces a precision-based, analysis-driven approach to chemical treatment in reservoir management. The reduced chemical volumes and decreased intervention frequency underscore its sustainability impact and operational efficiency, establishing it as a transformative solution for modern reservoir management. Scalable and cost-effective, this technology maximizes treatment efficacy while supporting a shift toward more sustainable industry practices

**Keywords:** Rigless chemical injection, capillary injection technology, precise treatment placement, reservoir management, corrosion prevention, scale control, environmental sustainability, operational efficiency

## Introduction

Chemical injections are a crucial component in the management of oil and gas reservoirs. These injections aim to combat detrimental downhole conditions that can impede production efficiency and safety. Across various applications and geographic settings, chemicals are used to inhibit corrosion, manage scale deposition, and mitigate liquid loading (Gregg and Lerbscher 2004; Pate and Stanley 2011). Effective chemical management is vital for maintaining the integrity of production equipment and optimizing overall reservoir performance. The adoption of chemical injection methodologies has significantly evolved over the past decades (Matthews and Frazier 2014). Chemical treatments are applied to curb negative impacts within a reservoir. Corrosion, scale buildup, and liquid loading can drastically reduce production rates and increase maintenance costs. By systematically injecting the appropriate chemicals, operators can maintain optimal conditions and extend the operational lifespan of production equipment (Kara et al. 2022).

The primary focus has shifted toward enhancing chemical efficiency while adhering to environmental regulations. Traditional batch injection methods, despite their historical use, can lead to various operational challenges. Conventional methods often result in uneven chemical distribution throughout a wellbore. This unevenness can lead to ineffective treatments, wherein chemicals may fail to reach critical areas that require treatment action, resulting in suboptimal performance and the necessity for additional treatments, and such inefficiencies increase chemical consumption and operational costs (Stewart-Liddon et al. 2014). The environmental implications of traditional chemical injection methods are receiving heightened scrutiny as the industry adopts sustainable practices. Excessive chemical usage raises the risk of spills and leaks, contributing to a negative environmental impact. Furthermore, the disposal of unused chemicals poses additional risks, making the reduction of chemical waste vital for regulatory compliance and public acceptance. Operational expenditures tied to traditional cyclic injection methods can be substantial. These methods necessitate ongoing monitoring and manual adjustments, contributing to elevated labor and overhead costs. As operational budgets tighten, there is increasing pressure to explore more efficient and environmentally responsible solutions for chemical treatments.

## Capillary Injection Technology for Chemical Treatment: An Overview

Capillary injections represent an engineering advancement in chemical treatment methodologies within the oil and gas industry. Leveraging the principles of capillary action, this technique allows for the precise and efficient delivery of chemicals to targeted zones to address critical challenges such as corrosion, scale formation, and liquid loading. The ability to inject chemicals accurately reduces waste, lowers costs, and minimizes the environmental impact compared to traditional bulk injection methods. As engineering teams continue to seek efficient and sustainable approaches to maximize reservoir performance, capillary injections stand out as a transformative solution that combines engineering excellence with operational practicality. The primary advantage of capillary injection is its ability to deliver chemicals precisely to targeted zones, thereby maximizing treatment effectiveness while minimizing environmental impact and operational costs (Andagoya Carrillo et al. 2022).

At its core, capillary injections utilize the natural phenomenon of capillary action, which refers to the ability of liquids to rise or travel within narrow spaces against gravity. In practical terms, capillary strings with typically small-diameter tubes facilitate the movement of chemicals into the treatment spot. The design and engineering of these strings are critical to the effectiveness of the capillary injection approach, with standard diameters ranging from 0.25 to 1 inch to allow optimal fluid flow while maintaining sufficient capillary pressure for effective chemical delivery (Pate and Stanley 2011). This capability is particularly valuable in situations where traditional bulk injections might result in excessive chemical use without adequately addressing localized issues. Capillary injection technology has proven especially effective in mitigating common production problems, enhancing operational efficiency and safety in oil and gas operations (Osorio, Chipuk, and Anderson 2017).

The applications of capillary injection are broad and varied, effectively addressing several common challenges encountered in oil and gas operations, as shown in Table 1. Liquid loading involves the pinpoint delivery of foaming agents and surfactants to effectively unload fluids from production systems. This targeted approach enhances the wells' efficiency by ensuring that accumulated liquids are effectively removed, allowing for increased gas flow and overall production optimization. In the domain of corrosion and scale control, the injection of precisely engineered corrosion inhibitors plays a crucial role in managing these challenges right at their source through delivering the right chemicals directly to areas susceptible to corrosion and scaling. Engineering teams can significantly extend the lifespan of their equipment, reduce maintenance costs, and maintain optimal production rates. For salt control, targeted and customized production chemical solutions are essential in stopping salt build-up precisely at the perforations, and this strategic engineering approach prevents the formation of salt deposits that can impede fluid flow and damage production equipment, ensuring that wells operate smoothly and efficiently. Paraffin control is achieved through the delivery of chemicals to the precise depth just beneath where paraffin begins to form, so proactively managing paraffin deposition will minimize blockages and maintain consistent production, reducing the need for remedial operations and enhancing overall system reliability. Also, effective control of H<sub>2</sub>S and CO<sub>2</sub> is facilitated through specialized design, coupled with proper chemical and alloy selections. This approach ensures that the risks associated with these gases are minimized, thereby protecting equipment integrity and enhancing safety in production environments. Through utilizing chemical treatments injected directly into selected areas/locations, the engineering teams can enhance the lifespan of their assets, thereby reducing maintenance costs and downtime associated with repairs (Matthews and Frazier 2014).

**Table 1—Capillary Applications and Solutions within the Oil and Gas Industry**

| Well Problem                                     | Capillary Applications & Solutions                                                                                                                           |
|--------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Liquid Loading</b>                            | Precise placement of foaming agents and surfactants facilitates the removal of accumulated fluids from the wellbore.                                         |
| <b>Corrosion and Scale Control</b>               | Strategically injected corrosion inhibitors are formulated to prevent both corrosion and scale formation at their origin.                                    |
| <b>Salt Control</b>                              | Tailored chemical treatments are applied directly at the perforation zones to prevent salt accumulation.                                                     |
| <b>Paraffin Control</b>                          | Treatment chemicals are accurately delivered to the depth just below the paraffin formation point to inhibit buildup.                                        |
| <b>H<sub>2</sub>S and CO<sub>2</sub> Control</b> | Effective control of hydrogen sulfide and carbon dioxide is achieved through specialized system design and the selection of compatible chemicals and alloys. |

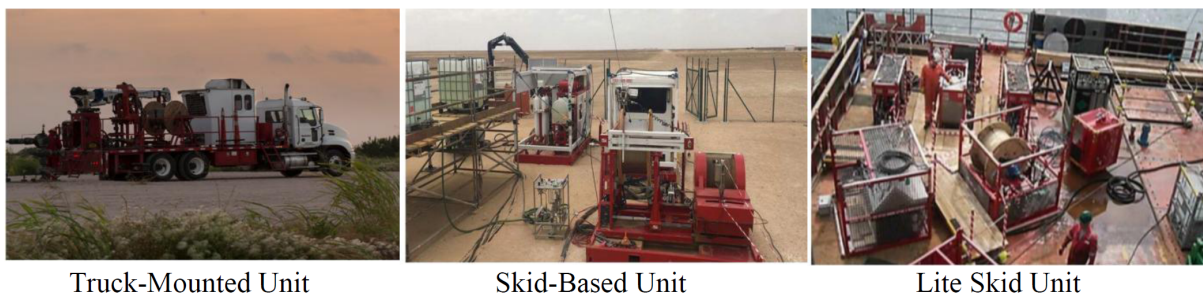
## Capillary Technology: Engineering and Installation Considerations

The successful implementation of a capillary injection system involves careful engineering design and installation. The capillary string must be designed to ensure the materials can withstand harsh downhole

conditions, including high temperatures and corrosive environments. Additionally, the diameter and length of the capillary string vary depending on the specific well conditions and the necessary treatment depth.

Installation begins with site assessment and well preparation. Prior to deploying the capillary string, engineers conduct thorough assessments of the well conditions, including pressure, temperature, and historical production data. This information informs decisions regarding the optimal chemical formulations to inject and the most effective installation strategy. During the installation process, the capillary string is carefully positioned within the wellbore to ensure optimal placement at the target interval. This usually involves coiled tubing techniques for quick installation and reduced operational disruptions.

Capillary technology allows for the insertion of a capillary string within the existing production tubing, enabling precise chemical delivery without requiring a workover rig. To support various operational environments, several unit configurations are available, as shown in [Figure 1](#). Truck-mounted units are designed for easy mobilization across different onshore sites, offering flexibility and rapid deployment. Skid-based units are well-suited for onshore operations and can be integrated with slickline systems for enhanced functionality. For offshore applications, lite skid units provide a compact and lightweight solution, ensuring ease of transport and installation in space-constrained environments.



**Figure 1—Capillary Technology Surface Unit**

To enhance automation and promote sustainable operations, several advanced surface equipment components are utilized. These include a chemical injection skid equipped with electric or pneumatic injection pumps, supported by data acquisition systems for real-time monitoring and optimization. A solar panel power source ([Figure 2](#)) ensures uninterrupted power supply, contributing to reduced carbon emissions and aligning with sustainability goals ([Brunet-Coy, Gouayat, and Lesage 2024](#)). This setup represents a one-time capital expenditure that delivers long-term operational efficiency. The benefits of this system include continuous surveillance of the injection process, ensuring precise and efficient chemical delivery. It also enables data-driven decision-making, allowing for informed and proactive interventions. Additionally, further remote control and adjustment capabilities will minimize human exposure, enhance safety, and facilitate remote management of the injection process.



**Figure 2—Solar Panel Power Source**



The capillary injection string method utilizes a small-diameter string, typically ¼ inch or 3/8 inch in diameter, which is installed internally or externally based on the engineering design for the chemical treatment application. This arrangement allows access to the tubing at a predetermined point within the well. This method proves to be an efficient and economical means of addressing various operational challenges while maintaining uninterrupted production. The capillary string facilitates the injection of various chemicals, including corrosion inhibitors, paraffin and hydrate solvents, H<sub>2</sub>S scavengers, CO<sub>2</sub> scavengers, foaming agents, and other chemicals for the treatment operations. The system comprises several key components (Matthews and Frazier 2014):

- Chemical injection sub: that is run into the tubing string, providing direct access to the internal diameter (ID) at the required injection point.
- Chemical injection valve: At the lowest injection point, the injection valve is attached to control the flow of injected chemicals and functions as a check valve for the capillary string. The valve's opening pressure is set before installation by adjusting the internal spring tension, ensuring it remains closed and prevents backflow of well fluids into the capillary string.
- Capillary string: It is installed concurrently with the production tubing using a derrick-mounted sheave and a deck-mounted spooling unit, and the string is secured to the tubing using either bands or stainless-steel cannon clamps, with the choice of band type based on well deviation.
- Wellhead exit: An extended neck tubing hanger and ported adapter flange are typically utilized for both onshore and offshore scenarios, and this setup can accommodate multiple injection ports when additional injection lines are required, allowing for chemical pumping via a standard pneumatic injection pump.
- Surface injection manifold: Surface manifold equipped with a micron filter to prevent contaminants from entering the downhole environment and includes a pressure gauge to monitor injection pressure, record values if necessary, and indicate the integrity of the capillary string. The manifold functionality also allows for estimating bottom hole pressure and features an anti-siphon device to mitigate suction effects generated by the hydrostatic pressure of the fluid in the capillary string opposing bottom hole pressure. The manifold is rated as pressures of 5,000 psi or 10,000 psi.
- Capillary injection tubing anchor: While the capillary injection method has been successfully implemented for various applications, including plunger lift installations, it presents a notable limitation in rod pump wells, specifically its inability to inject chemicals below tubing devices such as tubing anchors. This limitation is critical since it affects the capacity to reduce pump and rod wear due to corrosion. To address this issue, a capillary injection tubing anchor has been developed. This anchor device allows the capillary string to pass through it, enabling chemical injections directly at the bottom of the well. The system includes a 5/8-inch control section encased within the anchor's body, permitting the passage of a ¼-inch or 3/8-inch capillary string to an injection sub and valve positioned below as needed. The advantages of this device include:
  - Ensuring chemicals are injected into the production stream before entering the rod pump, protecting the pump and the entire rod and tubing system.
  - Enhancing pump efficiency and reducing wear on tubing and rods.
  - Commonly employed at depths exceeding 5,000 feet, though applicable in shallower, high-volume wells.
  - Allowing gas to flow into the annulus.
  - Incorporating a J-latch setting mechanism that facilitates installation by a simple ¼ turn of the tubing string at the tool, followed by the application of necessary tension according to well geometry.

The injection process can be executed as continuous or batch injection depending on the specific operational needs (Esquivia et al. 2023). Continuous injections allow for ongoing treatment of challenges such as corrosion and scale buildup, while batch injections may be employed for specific interventions and according to operational strategy.

The injection system requires continuous monitoring to ensure effective operation. Post-treatment evaluations are also essential, allowing engineering teams to assess the effectiveness of the chemical injection by reviewing parameters such as production rates, corrosion levels, and scaling tendencies. Utilizing sensors and automated control systems, engineering teams can continuously track metrics such as pressure, flow rates, and chemical concentrations. This capability enables technical adjustments in response to the changing conditions, ensuring efficient chemical delivery and treatment. Moreover, the ability to monitor downhole conditions accurately enhances safety by ensuring that operations remain within designed pressure and temperature limits, preventing potential failures.

## Field Applications and Results

The research outlines a structured engineering workflow (Figure 3) implemented for field application for the capillary injection technology. The process begins with problem identification and analysis, where engineers assess downhole challenges such as corrosion, scaling, and liquid loading. This step is critical for defining the scope of intervention and relies heavily on accurate data collection. Parameters such as corrosion rates, scale history, temperature, and pressure are gathered to build a comprehensive understanding of well conditions.

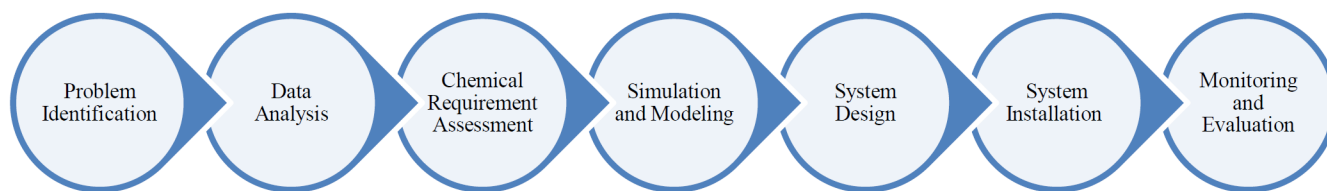


Figure 3—Structured Engineering Workflow

Following data collection, the chemical requirement assessment phase evaluates the types and quantities of chemicals needed. This includes reviewing historical chemical usage, compatibility with well materials, and leveraging analytics to tailor dosages for specific well environments. Next, simulation and modeling are critical in designing the injection strategy and tools such as multiphase flow modeling simulate fluid behavior and chemical interactions within the reservoir. These simulations inform decisions on chemical dosing, timing, and placement depth, ensuring optimal treatment efficiency.

The insights from modeling guide the design of the capillary injection system, including the capillary string and a customized injection valve. The capillary string is engineered for durability and precision, while the injection valve ensures accurate flow control and seamless integration into the wellbore.

The implementation phase involves the rigless installation of the capillary system without the need for a workover rig, significantly reducing costs and operational downtime. The capillary string is deployed based on simulation metrics, and the injection valve is activated to begin chemical delivery. Monitoring is integral to the system's success, and sensors and analytics tools continuously track injection performance, allowing for immediate adjustments to optimize chemical usage and ensure safety. This feedback loop enhances operational control and treatment effectiveness.

### Foaming Agent Capillary Injection for Liquid Loading Mitigation

Liquid loading in mature gas wells often leads to hydrostatic pressure buildup, restricting flow and necessitating intervention to restore production. This discussion highlights the application of capillary

injection technology to stabilize two liquid-loaded examples. Both examples experienced liquid loading, causing erratic production and declining rates. Conventional methods, such as workovers or velocity strings, were deemed unsuitable due to high costs, operational complexity, and risks of formation damage.

- **Diagnostic Analysis:** Well performance analysis software evaluated liquid-loading severity and optimized surfactant injection parameters.
- **Capillary System Deployment:** A truck-mounted capillary unit, rigged up with an injector head, spooler, and pressure control equipment, installed the capillary strings to target depths.
- **Foaming Agent Injection:** A tailored surfactant blend was injected continuously to reduce hydrostatic pressure. A defoamer was added at the surface to stabilize the flow for processing.
- **Operational Safety:** pressure control equipment ensured safe deployment, with capabilities to shear the capillary string during emergencies.

Capillary injection technology employs small-diameter tubing (0.25-inch OD) run alongside production tubing to deliver foaming agents directly into the wellbore (Figure 4).



Figure 4—Foam-Lift and Chemical-Injection System Field Application

This method eliminates the need for well kill operations or rig-based interventions, ensuring continuous production and minimizing downtime. The foaming agent lowered hydrostatic pressure by reducing fluid density, enabling gas to overcome critical velocity thresholds. Continuous injection maintained stable flow regimes, while real-time monitoring ensured optimal surfactant dosage. Post-intervention pressure surveys confirmed sustained bottomhole pressure reduction, validating liquid unloading.

Capillary injection technology effectively mitigated liquid loading in both wells, stabilizing production and extending operational life. The implementation of the rigless capillary injection technique yielded substantial production improvements in both wells. Well 1 experienced a 31–53% increase in gas output, while Well 2 reported a 31–84% production increase. Notably, both wells transitioned from erratic to stable flow patterns, clearly indicating successful liquid unloading and enhanced reservoir performance. The solution provided a cost-effective, low-risk alternative to conventional methods, aligning with sustainable production goals.

Additional field trials conducted across multiple candidates demonstrated the technique's value. Wells previously affected by severe corrosion and scaling showed a 65% reduction in chemical usage and a 52% decrease in operational costs. These improvements were attributed to targeted chemical delivery and reduced manual intervention. The post-trial evaluation confirmed the technique's scalability and environmental benefits. It aligns with industry sustainability goals by reducing the environmental footprint and ensuring regulatory compliance.

## Results in Focus: The Integration of Engineering, Sustainability, and Economic Approaches

This research outlines the significant benefits of advanced chemical treatment systems in enhancing oil and gas production while emphasizing engineering effectiveness, environmental responsibility, and economic practicality. Targeted treatment strategies have been shown to improve production rates by addressing specific issues, such as scale formation and corrosion, through accurate chemical applications.

Environmental responsibility is a key aspect of these advanced systems, which aim to reduce the operational footprint of oil and gas production. Efficient chemical applications will reduce waste generation significantly, aligning with regulatory requirements and corporate sustainability goals.

From an economic standpoint, the use of continuous chemical delivery systems reduces the necessity for costly interventions. This approach lowers operational costs related to workovers and minimizes production downtime, thereby improving overall operational performance. The prolonged asset life achieved through effective management of production challenges is another important benefit and contributing to the long-term durability of assets.

The capillary injection approach integrates engineering principles with environmental and economic factors, supporting the responsible extraction of hydrocarbons.

## Conclusion

- This research presented the capillary injection as a transformative approach to overcoming significant challenges in oil and gas production. The capillary action principles provided precise delivery of chemical treatments that helped to effectively manage corrosion, control scale, and mitigate liquid loading. The successful engineering and installation of capillary injection systems necessitate careful selection of materials, comprehensive monitoring, and strategic operational planning to optimize performance.
- The advantages of capillary injections extend well beyond immediate operational gains. This technique contributes to the long-term durability of production assets by enhancing integrity, reducing maintenance expenses, and decreasing the need for frequent interventions. Additionally, the minimized volume of chemicals required not only curtails operational costs but also lessens environmental impact, supporting industry commitments to sustainable practices. By implementing capillary injection technology, oil and gas operators enhance efficiency, improve safety, and ensure compliance with environmental regulations, leading to superior overall performance.
- As industry increasingly prioritizes sustainable and efficient methods for resource management, capillary injection emerges as a prominent solution that combines engineering innovation with practical applications. Adopting this method enables engineering teams to optimize production performance, decrease environmental footprints, and extend asset lifespans. With continuous research and development in capillary injection technology, further innovations in capillary injection can be anticipated that will deliver even greater efficiencies and capabilities, shaping the future of oil and gas production.

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